



ML KEY TERMS

Introduction to Machine Learning in 50 terms

Decode machine learning—one term at a time.

1

Algorithm

A step-by-step procedure or formula for solving a problem. In machine learning, algorithms are used to create models that can make predictions or decisions based on data.

2

Accuracy

A metric used to evaluate classification models, representing the ratio of correctly predicted instances to the total instances.

3

Anomaly Detection

The process of identifying rare or unusual data points that deviate significantly from the majority of the data. Useful in fraud detection and network security.

4

Bias

In machine learning, bias refers to the error introduced by approximating a real-world problem, which may be too complex, with a simplified model.

5

Classification

A type of supervised learning where the goal is to predict discrete labels or categories for given inputs, such as determining whether an email is spam or not.

6

Clustering

An unsupervised learning technique used to group similar data points together into clusters based on their features, without predefined labels.

7

Confusion Matrix

A table used to evaluate the performance of a classification model by showing the number of true positives, true negatives, false positives, and false negatives.

8

Cross-Validation

A technique for assessing how well a model generalizes to new data by splitting the dataset into training and testing subsets multiple times.

9

Data Preprocessing

The steps taken to clean, normalize, and transform raw data before feeding it into a machine learning model to improve accuracy and performance.

10

Deep Learning

A subset of machine learning involving neural networks with many layers (deep neural networks) that can automatically learn features from data.

11

Feature Engineering

The process of creating new input features or modifying existing ones to improve model performance. It involves selecting, transforming, and constructing features.

12

Feature Selection

The process of choosing a subset of relevant features for model building, aiming to reduce dimensionality and improve performance.

13

Gradient Descent

An optimization algorithm used to minimize the cost function by iteratively adjusting the model parameters in the direction of the negative gradient.

14

Hyperparameter

A parameter set before the learning process begins that controls the learning process itself, such as learning rate, number of trees in a forest, or depth of a neural network.

15

K-Nearest Neighbors (KNN)

A classification algorithm that assigns a label based on the majority vote of the nearest k data points in the feature space.

16

Label

The output or target value that the model is trying to predict. In supervised learning, labels are provided in the training data.

17

Learning Rate

A hyperparameter that controls how much to adjust the model parameters in each iteration of gradient descent. It affects the convergence speed and stability.

18

Logistic Regression

A classification algorithm used to predict the probability of a binary outcome, such as whether an email is spam or not.

19

Loss Function

A function that measures the difference between the predicted values and the actual values. It guides the optimization process by quantifying how well the model performs.

20

Model

A representation of a learned relationship from training data that can make predictions or decisions based on new, unseen data.

21

Overfitting

A situation where a model learns the training data too well, including noise and outliers, which leads to poor performance on new, unseen data.

22

Precision

A metric used to evaluate classification models, representing the ratio of true positive predictions to the total predicted positives.

23

Recall

A metric used to evaluate classification models, representing the ratio of true positive predictions to the total actual positives.

24

Regularization

Techniques used to prevent overfitting by adding a penalty to the loss function for large or complex model parameters, such as L1 and L2 regularization.

25

Regression

A type of supervised learning where the goal is to predict continuous values or quantities, such as predicting house prices based on features.

26

ROC Curve

A graphical representation of a classification model's performance across different thresholds, plotting the true positive rate against the false positive rate.

27

Support Vector Machine (SVM)

A supervised learning algorithm used for classification and regression tasks, which finds the optimal hyperplane that separates different classes.

28

Supervised Learning

A type of machine learning where the model is trained on labeled data, with input-output pairs, to learn the relationship between them.

29

Unsupervised Learning

A type of machine learning where the model is trained on unlabeled data, aiming to uncover hidden patterns or structures in the data.

30

Validation Set

A subset of data used to tune hyperparameters and assess the model's performance during training, helping to prevent overfitting.

31

Variance

A measure of how much the predictions of a model change with different subsets of the training data. High variance indicates overfitting.

32

Weights

Parameters in a machine learning model that are learned from the training data and determine the importance of each feature in making predictions.

33

Epoch

One complete pass through the entire training dataset. Multiple epochs are often required to train a model effectively.

34

Ensemble Learning

A technique that combines multiple models to improve overall performance, often by averaging predictions or voting.

35

Principal Component Analysis (PCA)

A dimensionality reduction technique that transforms data into a lower-dimensional space by finding the directions (principal components) of maximum variance.

36

Autoencoder

A type of neural network used to learn efficient representations of data by compressing it into a lower-dimensional space and then reconstructing it.

37

Decision Tree

A supervised learning algorithm that splits data into branches based on feature values to make predictions, visualized as a tree-like structure.

38

Random Forest

An ensemble learning method that combines multiple decision trees to improve performance and reduce overfitting, by averaging their predictions.

39

Neural Network

A computational model inspired by the human brain, consisting of interconnected nodes (neurons) organized in layers, used for complex pattern recognition tasks.

40

Support Vector Machine (SVM)

An algorithm that finds the optimal hyperplane to separate different classes in a high-dimensional space, used for both classification and regression tasks.

41

Bagging (Bootstrap Aggregating)

An ensemble technique that improves model stability and accuracy by training multiple models on different subsets of the data and combining their predictions.

42

Boosting

An ensemble technique that combines multiple weak learners to create a strong learner by sequentially training models and focusing on the errors of previous models.

43

Dimensionality Reduction

Techniques used to reduce the number of features or dimensions in the data, making it easier to visualize and analyze, while preserving essential information.

44

Generative Model

A model that learns to generate new samples from the same distribution as the training data, often used for data synthesis and augmentation.

45

Latent Variable

A variable that is not directly observed but inferred from the model, helping to capture underlying patterns or structures in the data.

46

Markov Chain Monte Carlo (MCMC)

A method for sampling from probability distributions to approximate complex distributions, often used in Bayesian statistics.

47

Naive Bayes

A classification algorithm based on Bayes' theorem with the assumption that features are independent given the class label, used for probabilistic classification.

48

Reinforcement Learning

A type of machine learning where an agent learns to make decisions by receiving rewards or penalties based on its actions in an environment.

49

Semi-Supervised Learning

A learning approach that uses a combination of a small amount of labeled data and a large amount of unlabeled data to improve model performance.

50

Tuning

The process of adjusting model hyperparameters to optimize performance, often involving techniques like grid search, random search, or Bayesian optimization.